

How information flows

one value — “striatum” — transformed into a grounded, reasoner-verified, computable axiom

INPUT

“striatum”

a raw string in the CurationRequest (location_hint)

↓ *ground · ols_search* → EBI OLS4 (cache-first)

GROUNDING

UBERON:0002435

a TermMatch — label ‘striatum’, score 1.00, source EBI OLS4

↓ *assemble · draft_definition*

AXIOM

part of some UBERON:0002435

a differential in the OWL class expression (BFO:0000050)

↓ *verify · ROBOT · ELK* vs *cl-base.owl*

VERIFICATION

NOVEL — under interneuron

ELK over the whole CL: coherent, subsumes its genus, no duplicate

↓ *emit · dossier renderers*

OUTPUT

KGCL · ROBOT/OWL · SSSOM · KG triples

every emitted format carries UBERON:0002435

CONTROL

the A1-style agent loop decides the order and self-corrects: **retrieve** → **plan** → **execute** → **self-correct** → **critique**

(if the genus doesn’t ground, it re-derives from the description and retries once).

PROVENANCE

every hop is auditable:

- each object is a dataclass carrying its source
- each step is logged as a Step (inputs + output summary) in dossier.trace
- the LLM (optional) only PLANS + PHRASES — it never mints an identifier

Three flows run together: CONTROL chooses what runs, DATA is grounded then verified, PROVENANCE is carried at every step.